
Expectation and Midterm 1

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Probability and Statistics for Engineers Fall 2026

Monday, October 12	Expectation, linearity, and indicators
Wednesday, October 14	Review through the binomial distribution
Friday, October 16	Midterm 1

Course support and deadlines

Ed Discussion: questions, announcements, and private messages. **No Canvas Inbox.**

Open Ed from Canvas All times Pacific

Full contact details and meeting links

Instructor / TAs	Office hours (Zoom)	Optional, ungraded discussions
Antonio de Leon	Wed 4–5 p.m.	—
Jeannine Romero	Tue / Thu 10–11 a.m.	01A: Fri 9:20–10:25 a.m. R Carson Acad 240 (in person)
Ehsan Mirafzali	Mon 3–5 p.m.	01B: Fri 10:40–11:45 a.m. (Zoom)
Seokjun Choi	Wed 3–4 p.m. 5:05–6:05 p.m.	01C: Wed 4–5:05 p.m. (Zoom)

Upcoming deadlines

HW and quizzes: **10 p.m. Pacific**

HW 2	Mon, Oct 12	Gradescope
Quiz 3	Wed, Oct 14	Canvas
Midterm 1	Fri, Oct 16	During class; Classroom Unit 002

Midterm 1 covers required material through Oct 9; expectation is excluded.

HW: 24-hour grace period. Quizzes: no grace period. Approved accommodations apply.



Expectation and the examination boundary

Today

Compute a mean, derive linearity, and use indicators to obtain a count's mean.

Midterm 1

Required material through October 9.

Expectation, variance, and joint distributions are excluded.

Recall the model. Three independent fair coin tosses; X counts heads.

x	0	1	2	3
$p_X(x)$	1/8	3/8	3/8	1/8

Why does $X = 1$ have three times the probability of $X = 0$?

Expected value

Definition for a finite-valued random variable

$$\mu = \mathbb{E}[X] = \sum_{x \in S_X} x p_X(x), \quad S_X = \{x : p_X(x) > 0\}.$$

Each possible value receives its probability as a weight.

$$p_X(x) \geq 0, \quad \sum_{x \in S_X} p_X(x) = 1.$$

A useful check. If $a \leq X \leq b$, why must $a \leq \mathbb{E}[X] \leq b$?

X is a random variable; $\mathbb{E}[X]$ is a number determined by its distribution.

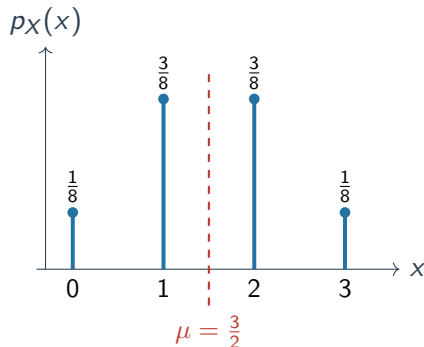
For countably many values, we assume $\sum_x |x| p_X(x) < \infty$. All required examples today have finitely many values.

Mean of a three-toss count

Values and their weights

x	0	1	2	3
$p_X(x)$	$1/8$	$3/8$	$3/8$	$1/8$

Write each contribution $x p_X(x)$, then add.



The mean is 1.5 heads, although the count is never 1.5.

An expectation summarizes a probability model. An observed average is calculated from data.

Grouping outcomes by value

On a finite sample space, let $q_\omega = P(\{\omega\})$. These weights need not be equal.

One mass combines outcome weights

Outcomes	$X(\omega)$
TTT	0
HTT, THT, TTH	1
HHT, HTH, THH	2
HHH	3

$$p_X(x) = \sum_{\omega: X(\omega)=x} q_\omega.$$

The three one-head outcomes each contribute $1 \cdot (1/8)$ to the mean.

Fair tosses: $q_\omega = 1/8$.

Both sums collect the same terms, grouped in different ways.

Linearity of expectation

For integrable X, Y and real constants a, b

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

We prove the finite-sample-space case. Both variables use the same outcome ω .

$$Z(\omega) = aX(\omega) + bY(\omega).$$

By the outcome representation,

$$\mathbb{E}[Z] = \sum_{\omega \in \Omega} Z(\omega)q_{\omega}.$$

Substitute the definition of Z and distribute the weights.

Integrable means $\mathbb{E}[|X|], \mathbb{E}[|Y|] < \infty$. This holds automatically for the finite models here.

Linearity on a finite sample space

We have

$$\mathbb{E}[aX + bY] = \sum_{\omega \in \Omega} [aX(\omega) + bY(\omega)] q_{\omega}.$$

Separate the two sums and identify each expectation.

Repeated application gives the finite-sum version

$$\mathbb{E}\left[\sum_{i=1}^n a_i X_i\right] = \sum_{i=1}^n a_i \mathbb{E}[X_i].$$

Constants and affine changes

A constant random variable has mean

$$\mathbb{E}[c] = \sum_{\omega \in \Omega} c q_{\omega} = c \sum_{\omega \in \Omega} q_{\omega} = c.$$

For integrable X and real constants a, b

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

The same coin experiment. Score $+1$ for a head and -1 for a tail. If X counts heads in three tosses, the total score is $Y = 2X - 3$.

Mean of an indicator

For an event A , its indicator records whether A occurs:

$$I_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

Value	0	1
Probability	$P(A^c)$	$P(A)$

Compute its probability-weighted average.

Indicator identity

$$\mathbb{E}[I_A] = P(A).$$

For $I \sim \text{Bern}(p)$, this gives $\mathbb{E}[I] = p$, including $p = 0$ and $p = 1$.

Mean of a binomial count

Let A_i be success on trial i , with $P(A_i) = p$, and put $I_i = I_{A_i}$.

$$N = I_1 + \cdots + I_n.$$

Use linearity and the indicator identity.

Two assumptions to distinguish

The mean is np whenever all $P(A_i) = p$.

Mutually independent Bernoulli trials give the distribution $N \sim \text{Bin}(n, p)$.

More generally, $\mathbb{E}[N] = \sum_i P(A_i)$ even with dependence or unequal success probabilities.

Sampling without replacement

A count with dependent indicators

A bag has three red balls and two blue balls. Draw two balls uniformly without replacement. Let N count red balls, $A_i = \{\text{draw } i \text{ is red}\}$, and $I_i = I_{A_i}$.

- 1 Express N in terms of I_1, I_2 and find its mean.
 - 2 Explain why the draws are dependent.
-
-
-
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Each draw has unconditional red probability $3/5$. The conditional probability changes.

Expectation: summary

Object or operation	Result
Discrete mean	$\mathbb{E}[X] = \sum_x x p_X(x)$
Finite linear combination	$\mathbb{E}[\sum_i a_i X_i] = \sum_i a_i \mathbb{E}[X_i]$
Indicator	$\mathbb{E}[I_A] = P(A)$
Count of events	$\mathbb{E}[\sum_i I_{A_i}] = \sum_i P(A_i)$

Use finite expectations. Linearity does not require independence.

Examination boundary

Today's expectation material is excluded from Midterm 1.

Midterm 1 scope

Required material through October 9

Events and counting; conditional probability, total probability, and Bayes; independence; random variables, PMFs and CDFs; Bernoulli and binomial laws.

Excluded

Expectation, variance, and joint distributions.

Before a calculation Identify the experiment, the event or random variable, and the assumptions.

Retrieval: in a coin experiment, how do an outcome, the event $\{X = 2\}$, and the variable X differ?

Roles and committees

A student organization has 12 members. Select three different people.

Three distinct offices

President, vice-president, treasurer.

A committee of three

There are no distinct roles.

The population and selection size are the same. Only the role of ordering changes.

Four items in seven locations

Four labeled items independently choose among seven labeled locations, uniformly.

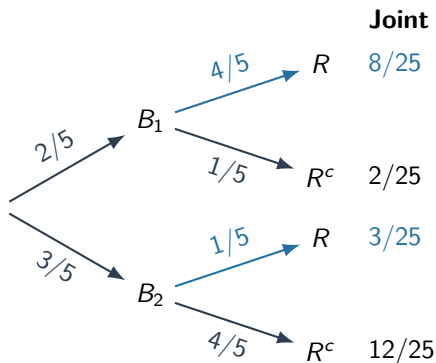


Find the probabilities of no collision and of at least one collision.

Independent uniform choices make the 7^4 labeled placements equally likely.

Source probabilities and red outcomes

Select source B_1 with probability $2/5$, otherwise B_2 . A result is red with probability $4/5$ from B_1 and $1/5$ from B_2 .



The branch probabilities are conditional

$$P(R \mid B_1) = \frac{4}{5}, \quad P(R \mid B_2) = \frac{1}{5}.$$

Explain one path, then add the red paths.

The sources form a partition. Red can occur through either disjoint source event.

Posterior probability

The result is red. What is the probability that source B_1 was selected?

Source	Prior	Red given source	Joint red contribution
B_1	$2/5$	$4/5$	$8/25$
B_2	$3/5$	$1/5$	$3/25$
Total			$11/25$

Restrict to red outcomes and renormalize. State the denominator's meaning.

$P(R \mid B_1) = 4/5$ answers a different conditional question.

Event operations and independence

Suppose A, B are independent, with $P(A) = 1/3$ and $P(B) = 1/4$.

Find both, at least one, and neither.

Independent

$$P(A \cap B) = P(A) P(B).$$

Disjoint

$$A \cap B = \emptyset.$$

Disjoint events of positive probability cannot be independent.

If $P(B) > 0$, independence is equivalent to $P(A | B) = P(A)$. A zero-probability event is independent of every event under the product definition.

Outcomes, values, and masses

$\Omega = \{1, \dots, 10\}$, with each outcome having probability $1/10$.

Find $\{X = 0\}$ and the PMF.

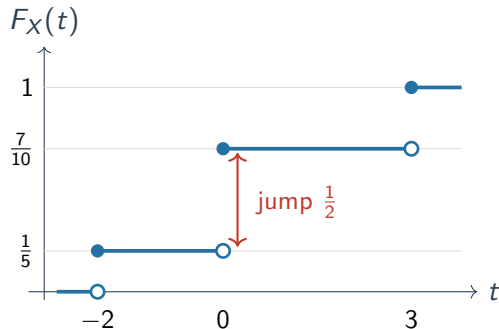
Outcomes	$X(\omega)$
1, 2	→ -2
3, 4, 5, 6, 7	→ 0
8, 9, 10	→ 3

PMF check

The positive-mass values are $-2, 0, 3$. Set the PMF to zero elsewhere; its masses must sum to one.

Masses and CDF jumps

$$F_X(t) = P(X \leq t), \quad t \in \mathbb{R}.$$



x	-2	0	3
$p_X(x)$	$\frac{1}{5}$	$\frac{1}{2}$	$\frac{3}{10}$

Find $P(X = 0)$, $P(X < 0)$, $P(X \leq 0)$,
and $P(-2 < X \leq 3)$.

$$P(a < X \leq b) = F_X(b) - F_X(a), \quad a < b.$$

A PMF mass is a CDF jump. The closed dot gives the CDF value at a jump.

Binomial probabilities

Eight independent test runs each detect a particular defect with probability $1/4$. Let N count detections. This is a hypothetical model.

$$N \sim \text{Bin}(8, 1/4), \quad N \in \{0, 1, \dots, 8\}.$$

Exactly two detections

At least one detection

Check fixed n , two outcomes per trial, common p , and mutual independence.

Pairwise independence alone does not justify the probability of a full pattern. Exact expressions suffice.

Review: final checks

Check	What to identify
Model	Which outcomes are equally likely? Is independence given?
Event or variable	What precisely is being counted or measured?
Counting	Do order, labels, and replacement matter?
Conditioning	Which event defines the denominator? Is its probability positive?
Distribution	What is the support? Do the masses sum to one?
CDF	Which endpoints are included? Is the request a mass or a cumulative probability?

Finish with the probability of the requested event. It must lie in $[0, 1]$.

Midterm 1

Examination meeting

Friday, October 16, 2026

55 minutes of working time within the scheduled class period.

Scope

Required material through October 9: events, counting, conditioning, independence, random variables, PMFs/CDFs, and Bernoulli/binomial laws.

Excluded

Expectation, variance, and joint distributions.

Materials

Instructor-provided formula sheet and a basic non-programmable calculator. No other notes, communication, or electronic resources.

Date follows the current syllabus. Approved accommodations are handled separately.

Optional: expectation of a function

For finite-valued X and a real-valued function g , put $Y = g(X)$. Grouping inputs that give the same output gives

$$\mathbb{E}[Y] = \sum_y y \sum_{x:g(x)=y} p_X(x) = \sum_x g(x)p_X(x).$$

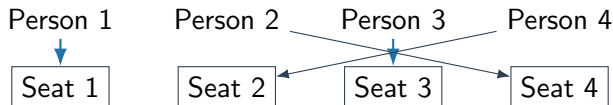
The probabilities stay attached to the original inputs.

x	0	1	2	3
$p_X(x)$	1/8	3/8	3/8	1/8
$g(x) = x^2$	0	1	4	9

For countably supported X , assume $\sum_x |g(x)|p_X(x) < \infty$. This extension is excluded from Midterm 1 and will be revisited before variance.

Optional: random matching

There are $n \geq 1$ labeled seats and n people, each with an assigned seat. Choose one of the $n!$ seating permutations uniformly. Let M count correct matches.

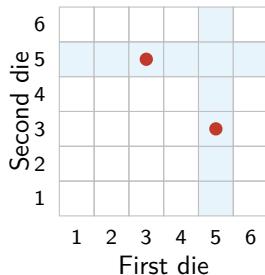


Let I_i indicate that person i gets seat i . Find $\mathbb{E}[M]$.

The diagram shows one seating of four people. The mean calculation covers all permutations.

Optional: changed conditioning

Roll two independent fair dice. Let S be their sum and B the event that at least one die shows 5.



Compare three observations

- ① At least one die shows 5.
- ② The first die shows 5.
- ③ Exactly one die shows 5.

What is the probability of $S = 8$ under each observation?

Count favorable outcomes inside the conditioning event.

References for this week

DeGroot and Schervish

Probability and Statistics, fourth edition.

- §4.1, pp. 207–209: discrete expectation and Bernoulli example.
- §4.2, pp. 217–221: affine changes, linearity, and sampling means.

Mor Harchol-Balter

Introduction to Probability for Computing, supplied 2024 version.

- §4.1, pp. 58–59: opening expectation definition and examples.
- §4.2, pp. 63–64: linearity and the binomial mean.
- Example 4.12, pp. 64–65: optional random matching.

Selections only. The lecture gives a finite-outcome proof in place of the books' joint-law or continuous arguments. Review references are the current Weeks 1–2 slides.

Additional source: Sheng Jiang, STAT 131 Notes, July 18, 2024, selected statements on pp. 48–51. Examples adapt Antonio de Leon's earlier STAT 131 sources.